

The empirical recurrence for OEIS A250947: recovering a conjecture that is not written down

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Abstract

OEIS A250947 records “Empirical recurrence of order 60”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 284$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 60, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 60 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A250947 is “Number of $(5+1)X(n+1) 0..2$ arrays with nondecreasing $x(i,j)+x(i,j-1)$ in the i direction and nondecreasing absolute value of $x(i,j)-x(i-1,j)$ in the j direction.”. It has offset 1 and begins

4213, 37477, 289407, 2067987, 13550488, 82716842, 475951361, 2607327593, ...

The entry states:

Empirical recurrence of order 60 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 12:53:32, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 729$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 284$ of the 729, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 284$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 1458$ exact terms of the model returns order 60 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{60} c_i a(n-i),$$

c_1	19	c_{16}	−2309496818	c_{31}	−641909479976	c_{46}	−4646317152
c_2	−104	c_{17}	−1083577001	c_{32}	−221974610846	c_{47}	−4752009961
c_3	−243	c_{18}	9479986396	c_{33}	681766086565	c_{48}	1380125671
c_4	4437	c_{19}	−734097028	c_{34}	120672104468	c_{49}	1008704421
c_5	−8783	c_{20}	−29734405278	c_{35}	−597956771073	c_{50}	−314500624
c_6	−64340	c_{21}	13765456171	c_{36}	−30440763060	c_{51}	−165117042
c_7	285000	c_{22}	73244230168	c_{37}	434421799690	c_{52}	54087735
c_8	343134	c_{23}	−56806163987	c_{38}	−19833613724	c_{53}	20052794
c_9	−3832793	c_{24}	−143848714414	c_{39}	−261458881991	c_{54}	−6793152
c_{10}	1839220	c_{25}	153029202896	c_{40}	31579095946	c_{55}	−1696888
c_{11}	30894973	c_{26}	226790165632	c_{41}	130027846805	c_{56}	587488
c_{12}	−46195204	c_{27}	−309799608159	c_{42}	−23368977324	c_{57}	89056
c_{13}	−163832914	c_{28}	−286791410020	c_{43}	−53152944740	c_{58}	−31232
c_{14}	408090324	c_{29}	495636146217	c_{44}	12015830462	c_{59}	−2176
c_{15}	570233291	c_{30}	287689401444	c_{45}	17711226857	c_{60}	768

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 60, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $60 + 4$ terms removed returns order 60 as well, so $\deg q_{\min} = 60 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $60 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 60 that this sequence satisfies — and that family turns out to have exactly one member.

References

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